

Groups and Lie Algebras: Exercise Sheet 2

1. Simple groups

A group G is called *simple* if it has exactly two normal subgroups; the trivial subgroup $\{1\}$ and G itself.

- a) Let G be a simple group and H a group. Show that every group homomorphism from G to H is either an injection or the constant map at the identity of H . Show that any group G' satisfying this property must be simple.
- b) For which $n \in \mathbb{N}$ is $\mathbb{Z}/n\mathbb{Z}$ simple?
- c) Let G be a group and $N \subset G$ a maximal normal subgroup, i.e. $N \neq G$ and $N \subsetneq N' \subset G$ with N' normal implies $N' = G$. Show that G/N is simple.
- d) Show that a group with only two conjugacy classes is simple.
- e) Iterating the construction from c) we can assign to each finite group G a sequence of simple groups. Show explicitly for $G = \mathbb{Z}/12\mathbb{Z}$ that the simple groups appearing in this decomposition are independent of the choice of maximal normal subgroup (this is true for every finite group).
- f) The decomposition in e) is called the Jordan–Hölder decomposition of G . Show that there are non-isomorphic groups with the same Jordan–Hölder decomposition.

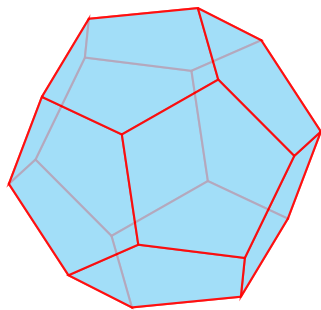
Informally, G can be reconstructed by ‘gluing’ the simple groups in its Jordan–Hölder decomposition together. Finite simple groups are the atoms of (finite) group theory. There exists a classification of all finite simple groups, which goes far beyond the scope of this lecture.

2. Spontaneous symmetry breaking

It can happen that a quantum mechanical system is in a ground state which does not preserve the symmetries of the system. In this case we say the symmetry is *spontaneously broken*. Consider a particle on a circle with Hilbert space $\mathcal{H} = L^2(\mathbb{R}/\mathbb{Z})$ and Hamiltonian $\mathbb{H} = \frac{1}{2}(-i\partial_x - \pi)^2$. Compute the eigenfunctions of $\mathbb{H}$ and show that $f(x) \mapsto e^{2\pi i x} f(-x)$ is a symmetry. What is the action of the symmetry when applied to the eigenfunctions?

3. Symmetries of the Dodecahedron

- a) Compute the size of the group of symmetries of the Dodecahedron



(tikz.net/dodecahedron/)

using the orbit stabilizer theorem.

- b) What is the size of the group of rotational symmetries?
- c) There are 5 cubes inside the Dodecahedron which can be rotated into each other and contain all its vertices. Show that there is an injection from the group of rotational symmetries to S_5 .
- d) Show that the image of the injection from c) is contained in the alternating group A_5 .
- e) Show that the group of symmetries of the Dodecahedron is $A_5 \times \mathbb{Z}_2$.

4. Cayley's theorem

Show that every finite group is isomorphic to a subgroup of a symmetric group. Hint: Use the action of a group on itself by left multiplication.